

The Mathieu Property for Compact Connected Lie Groups

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Abstract

Let G be a compact connected Lie group, let $\mathcal{R}(G)$ denote its algebra of representative functions, and let $\mathcal{I}_G(f) = \int_G f(g) \, dg$ be normalized Haar integration. We prove that $\ker \mathcal{I}_G$ is a Mathieu–Zhao subspace of $\mathcal{R}(G)$ if and only if G is a torus. More strongly, for every nonabelian compact connected Lie group G we construct $A, P, Q \in \mathcal{R}(G)$, with $A \geq 0$ and $A \not\equiv 0$, such that all pure moments of P vanish and the marked moments satisfy an exact Pascal-row identity

$$\mathcal{I}_G(Q^s P^m) = c_m \binom{m-1}{s-1} \mathcal{I}_G(A^{4m+s}) > 0 \quad (1 \leq s \leq m),$$

where $c_m = 4^m (m!)^2 / (2m+1)!$; the same moments vanish for $s > m$. The construction begins with a homogeneous, phase-balanced polynomial pair on $\mathbb{C}^2$. An exact coefficient identity on the Hopf sphere S^3 is transferred by orbit averaging to every compactly supported $SU(2)$ -invariant measure on $\mathbb{C}^2$. A highest-weight representation and a visible simple-root doublet produce such a measure on every compact simple Lie group. Phase balance gives descent through every central quotient, and pullback from an adjoint simple quotient handles arbitrary nonabelian compact connected groups. The proof is direct and independent of the implication from the Mathieu conjecture to the Jacobian conjecture. The torus direction is the theorem of Duistermaat and van der Kallen.

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1 Introduction

Mathieu formulated in [8] a moment conjecture for finite-type functions on compact connected Lie groups. In the language of Mathieu subspaces introduced and developed by Zhao [13], it asks whether the kernel of normalized Haar integration is a Mathieu subspace of the algebra of representative functions.

Duistermaat and van der Kallen proved the conjecture for tori [4]. Dings and Koelink studied the first nonabelian case through matrix coefficients [3]. Müger–Tuset and Zwart subsequently used explicit Euler- and KAK-type integration formulas to reduce the conjecture for $SU(2)$, $SU(N)$, $SO(N)$, $Sp(N)$, G_2 , and ultimately general compact connected Lie groups to moment or convex-support conjectures on abelian parameter spaces [9, 10, 14, 15, 16].

There are now two independent sources of counterexamples. First, an explicit counterexample for $SU(2)$ was given in [7]. Second, in July 2026 Alpöge announced an explicit three-dimensional counterexample to the Jacobian conjecture, credited in the announcement to Fable [1]. The announced map has constant Jacobian determinant -2 and is generically three-to-one [11]. Xena

reported that Lezeau had manually formalized the example and opened a pull request to the Formal Conjectures repository [12]; that pull request was merged on July 26, 2026 [6]. Postcomposing with the diagonal target automorphism $\text{diag}(-\frac{1}{2}, 1, 1)$ normalizes the Jacobian determinant to 1 while preserving noninjectivity. Adjoining $N - 3$ identity coordinates then gives a counterexample in every dimension $N \geq 3$. Since the Mathieu conjecture for $SU(N)$ implies the Jacobian conjecture in dimension N [8, 17], it follows that the Mathieu conjecture fails for $SU(N)$ for every $N \geq 3$. These results do not by themselves classify arbitrary compact connected groups. The purpose of the present paper is to give a direct, uniform classification, independent of the Jacobian route and without case-by-case analysis by Lie type.

Let $\mathcal{R}(G)$ be the algebra of representative functions on a compact group G , equivalently the finite linear combinations of matrix coefficients of finite-dimensional continuous complex representations. Let

$$\mathcal{I}_G : \mathcal{R}(G) \longrightarrow \mathbb{C}, \quad \mathcal{I}_G(f) = \int_G f(g) \, dg,$$

where Haar measure is normalized to have total mass one. We say that G has the *Mathieu property* if $\ker \mathcal{I}_G$ is a Mathieu subspace of $\mathcal{R}(G)$.

Our main result is the following complete classification.

Theorem 1.1 (Classification theorem). *Let G be a compact connected Lie group. Then the following conditions are equivalent:*

- (i) $\ker \mathcal{I}_G$ is a Mathieu–Zhao subspace of $\mathcal{R}(G)$;
- (ii) G is abelian;
- (iii) G is a torus.

The nonabelian direction is substantially stronger than the negation of eventual vanishing. We use the convention $\binom{r}{k} = 0$ when $k < 0$ or $k > r$.

Theorem 1.2 (Uniform nonabelian marker tower). *For every nonabelian compact connected Lie group G , there exist $A, P, Q \in \mathcal{R}(G)$ such that $A(g) \geq 0$ for every $g \in G$, $A \not\equiv 0$, and, for every integer $m \geq 1$,*

$$\mathcal{I}_G(P^m) = 0. \tag{1.1}$$

For every integer $s \geq 1$ one has

$$\mathcal{I}_G(Q^s P^m) = c_m \binom{m-1}{s-1} \mathcal{I}_G(A^{4m+s}), \quad c_m = \frac{4^m (m!)^2}{(2m+1)!} > 0. \tag{1.2}$$

Consequently, the marked moment is strictly positive for $1 \leq s \leq m$ and vanishes for $s > m$. In particular, the same fixed multiplier Q detects every positive power of P .

The proof has four steps. First, we establish an exact coefficient formula for a homogeneous, phase-balanced polynomial pair on the Hopf sphere S^3 . Second, we transfer the full coefficient identity by orbit averaging to arbitrary compactly supported $SU(2)$ -invariant measures on $\mathbb{C}^2$. Third, a fundamental representation and a highest-weight-one simple-root doublet produce such a measure on every simply connected compact simple group; phase balance makes the construction descend through every central quotient. Finally, every nonabelian compact connected group has an adjoint compact simple quotient. For the defining representations of $SU(n)$ and $Sp(n)$, Proposition 5.4 evaluates the entire marker tower in closed beta-function form.

The standard facts concerning representative functions, compact Lie groups, root subgroups, highest weights, and the associated $\mathfrak{sl}_2$ -modules used below may be found in [2, 5]. All moment calculations and all transfer arguments specific to the present construction are proved explicitly.

2 Mathieu subspaces and functoriality

Definition 2.1. Let A be a commutative unital $\mathbb{C}$ -algebra. A linear subspace $M \subseteq A$ is a Mathieu subspace if, whenever $f^m \in M$ for every $m \geq 1$, one has

$$hf^m \in M \quad \text{for every fixed } h \in A \text{ and all sufficiently large } m.$$

For a compact group G , we say that G has the Mathieu property if $\ker \mathcal{I}_G$ is a Mathieu subspace of $\mathcal{R}(G)$.

For compact groups, representative functions are precisely the finite-type functions appearing in Mathieu's original formulation. We record the closure and functoriality properties used later.

Lemma 2.2 (Representative functions). *Let G be a compact group.*

- (i) $\mathcal{R}(G)$ is a unital $*$ -subalgebra of $C(G)$.
- (ii) If f and h are matrix coefficients of representations ρ and σ , respectively, then fh is a matrix coefficient of $\rho \otimes \sigma$, and $\bar{f}$ is a matrix coefficient of the conjugate representation $\bar{\rho}$.
- (iii) If $\pi : G \rightarrow H$ is a continuous homomorphism of compact groups, then $f \circ \pi \in \mathcal{R}(G)$ for every $f \in \mathcal{R}(H)$.

Proof. The constant function 1 is the coefficient of the trivial representation. If $f(g) = \langle \rho(g)v, \lambda \rangle$ and $h(g) = \langle \sigma(g)w, \mu \rangle$, then

$$f(g)h(g) = \langle (\rho \otimes \sigma)(g)(v \otimes w), \lambda \otimes \mu \rangle.$$

Complex conjugation gives a coefficient of $\bar{\rho}$. Finally, $f \circ \pi$ is a coefficient of $\rho \circ \pi$ whenever f is a coefficient of ρ . Taking finite linear combinations proves the assertions. $\square$

Lemma 2.3 (Haar pushforward and pullback). *Let $\pi : G \twoheadrightarrow H$ be a continuous surjective homomorphism of compact groups. Then the pushforward of normalized Haar measure on G is normalized Haar measure on H . Consequently,*

$$\mathcal{I}_G(f \circ \pi) = \mathcal{I}_H(f) \quad (f \in C(H)). \quad (2.1)$$

In particular, a counterexample to the Mathieu property on H pulls back to a counterexample on G .

Proof. The pushforward measure is a left-invariant probability measure on H , hence is normalized Haar measure by uniqueness. Equation (2.1) follows. Lemma 2.2 shows that representative functions pull back to representative functions. $\square$

3 The universal Hopf pair

For a polynomial or formal power series $F(X)$, we write $[X^m]F(X)$ for the coefficient of X^m ; for a Laurent polynomial in u , we write CT_u for its constant term.

For $z = (z_0, z_1) \in \mathbb{C}^2$, define the phase-balanced quadratic polynomials

$$\begin{aligned} a &= |z_0|^2 + |z_1|^2, & \tau &= |z_0|^2 - |z_1|^2, \\ u &= 2z_0\bar{z}_1, & v &= 2z_1\bar{z}_0. \end{aligned} \quad (3.1)$$

They satisfy the Hopf relation

$$a^2 = uv + \tau^2. \quad (3.2)$$

Define

$$\boxed{\mathcal{P} = (a + u)(a^2v - (2a + u)\tau^2), \quad \mathcal{Q} = u.} \quad (3.3)$$

Both are polynomials in $z_0, z_1, \overline{z_0}, \overline{z_1}$. They are invariant under the common phase action $(z_0, z_1) \mapsto (\lambda z_0, \lambda z_1)$ for $|\lambda| = 1$, and their real homogeneities are

$$\mathcal{P}(rz) = r^8 \mathcal{P}(z), \quad \mathcal{Q}(rz) = r^2 \mathcal{Q}(z) \quad (r \geq 0). \quad (3.4)$$

The two structural constraints serve different later steps. Homogeneity makes each orbit-sphere moment a single radial power and is essential to the transfer argument in Section 4; phase balance makes all quantities invariant under scalar central characters and is essential to the descent argument in Lemma 5.5. Thus an inhomogeneous counterexample on the unit sphere need not survive this mechanism.

Let $S^3 = \{z \in \mathbb{C}^2 : a(z) = 1\}$ and let σ be normalized surface measure. Write

$$p = \mathcal{P}|_{S^3}, \quad q = \mathcal{Q}|_{S^3}.$$

For $m \geq 1$, set

$$c_m = \int_0^1 (1 - s^2)^m \, ds = \frac{1}{2} B\left(\frac{1}{2}, m + 1\right) = \frac{4^m (m!)^2}{(2m + 1)!} = \frac{2^m m!}{(2m + 1)!!}. \quad (3.5)$$

Here B denotes Euler's beta function. In particular, $c_m > 0$.

Theorem 3.1 (Hopf coefficient identity). *For every integer $m \geq 1$ and every polynomial $H \in \mathbb{C}[X]$,*

$$\int_{S^3} H(q(z)) p(z)^m \, d\sigma(z) = c_m [X^m] H(X) (1 + X)^{m-1}. \quad (3.6)$$

Proof. Use Hopf coordinates

$$z_0 = \sqrt{\frac{1+t}{2}} e^{i\alpha}, \quad z_1 = \sqrt{\frac{1-t}{2}} e^{i\beta}, \quad (3.7)$$

where $-1 \leq t \leq 1$ and $0 \leq \alpha, \beta < 2\pi$. Outside the two endpoint circles of measure zero, these coordinates give

$$d\sigma = \frac{1}{8\pi^2} \, dt \, d\alpha \, d\beta, \quad \tau = t, \quad u = \sqrt{1-t^2} e^{i(\alpha-\beta)}. \quad (3.8)$$

On S^3 , relation (3.2) becomes $uv + \tau^2 = 1$. Multiplying the definition of p by u gives the polynomial identity

$$up = (1 + u)(1 - (1 + u)^2 \tau^2). \quad (3.9)$$

Thus, away from the measure-zero set $u = 0$,

$$p = u^{-1}(1 + u)(1 - (1 + u)^2 t^2). \quad (3.10)$$

The preceding polynomial identity is the global justification for this localization. In particular, p is the restriction of a polynomial to the compact sphere and is bounded; the apparent pole in (3.10) is used only off the null set $u = 0$ and creates no integrability issue. Thus all integrals below are absolutely convergent.

For fixed $-1 < t < 1$, write $u = r_t e^{i\theta}$ with $r_t = \sqrt{1-t^2}$. Averaging over the phase $\theta = \alpha - \beta$ extracts the Laurent constant term in u : if $F(u) = \sum_j b_j(t) u^j$, then $(2\pi)^{-1} \int_0^{2\pi} F(r_t e^{i\theta}) d\theta = b_0(t)$. The (α, β) -integration contributes one half of this normalized phase average. The resulting constant term is even in t , so the factor $1/2$ cancels the factor 2 obtained by restricting from $[-1, 1]$ to $[0, 1]$. Consequently,

$$\begin{aligned} \int_{S^3} H(q) p^m d\sigma &= \frac{1}{2} \int_{-1}^1 \text{CT}_u \left(H(u) u^{-m} (1+u)^m (1 - (1+u)^2 t^2)^m \right) dt \\ &= \int_0^1 \text{CT}_u \left(H(u) u^{-m} (1+u)^m (1 - (1+u)^2 t^2)^m \right) dt \\ &= [X^m] H(X) (1+X)^m \int_0^1 (1 - (1+X)^2 t^2)^m dt. \end{aligned} \quad (3.11)$$

Define the odd polynomial

$$J_m(y) = \int_0^y (1-s^2)^m ds.$$

The substitution $s = (1+X)t$ in (3.11) gives

$$\int_{S^3} H(q) p^m d\sigma = [X^m] H(X) (1+X)^{m-1} J_m(1+X). \quad (3.12)$$

Moreover,

$$\begin{aligned} J_m(1+X) - J_m(1) &= \int_0^X (1-(1+y)^2)^m dy \\ &= \int_0^X (-y(2+y))^m dy \in X^{m+1} \mathbb{C}[X]. \end{aligned} \quad (3.13)$$

Since $H(X)(1+X)^{m-1}$ has no negative powers, the term in $X^{m+1} \mathbb{C}[X]$ contributes nothing to the coefficient of X^m in (3.12). Hence $J_m(1+X)$ may be replaced by $J_m(1) = c_m$, proving (3.6). $\square$

Corollary 3.2 (Pure moments and Pascal marker tower). *For every $m \geq 1$,*

$$\int_{S^3} p^m d\sigma = 0. \quad (3.14)$$

For every integer $s \geq 1$,

$$\int_{S^3} q^s p^m d\sigma = c_m \binom{m-1}{s-1}. \quad (3.15)$$

Thus the marked moments are exactly row $m-1$ of Pascal's triangle: they are positive for $1 \leq s \leq m$ and vanish for $s > m$. In particular,

$$\int_{S^3} qp^m d\sigma = c_m > 0. \quad (3.16)$$

Proof. Set $H = 1$ in Theorem 3.1. Since $(1+X)^{m-1}$ has degree $m-1$, the coefficient of X^m is zero. For $H(X) = X^s$, the required coefficient is

$$[X^{m-s}](1+X)^{m-1} = \binom{m-1}{s-1},$$

with the stated zero convention when $s > m$. $\square$

Remark 3.3 (Defect-one coefficient law). *Equation (3.6) is a one-step boundary law for a Pascal row. The unmarked extraction lies one degree beyond $(1+X)^{m-1}$ and vanishes, while multiplication by q^s moves the extraction to the coefficient of X^{m-s} . No additional finite-difference identity is needed in the proof.*

4 Transfer to radial measures

We identify $SU(2)$ with its defining unitary action on $\mathbb{C}^2$. This action is transitive on every sphere centered at the origin: for a unit vector $\zeta = (\zeta_0, \zeta_1)$, the matrix

$$\begin{pmatrix} \zeta_0 & -\overline{\zeta_1} \\ \zeta_1 & \overline{\zeta_0} \end{pmatrix} \in SU(2)$$

sends $(1, 0)^\top$ to ζ .

Theorem 4.1 (Universal radial transfer). *Let μ be a compactly supported finite positive Borel measure on $\mathbb{C}^2$ that is invariant under the defining action of $SU(2)$. Then, for every $m \geq 1$,*

$$\int_{\mathbb{C}^2} \mathcal{P}(z)^m d\mu(z) = 0. \quad (4.1)$$

For every integer $s \geq 1$,

$$\int_{\mathbb{C}^2} \mathcal{Q}(z)^s \mathcal{P}(z)^m d\mu(z) = c_m \binom{m-1}{s-1} \int_{\mathbb{C}^2} a(z)^{4m+s} d\mu(z). \quad (4.2)$$

Thus the marked moment vanishes for $s > m$. If $\mu(\mathbb{C}^2 \setminus \{0\}) > 0$, then

$$\int_{\mathbb{C}^2} a(z)^{4m+s} d\mu(z) > 0 \quad (m, s \geq 1),$$

and consequently

$$\int_{\mathbb{C}^2} \mathcal{Q}(z)^s \mathcal{P}(z)^m d\mu(z) > 0 \quad (m \geq 1, 1 \leq s \leq m). \quad (4.3)$$

Proof. For any μ -integrable function F , invariance of μ and Fubini's theorem give

$$\int_{\mathbb{C}^2} F(z) d\mu(z) = \int_{\mathbb{C}^2} \int_{SU(2)} F(kz) dk d\mu(z). \quad (4.4)$$

For fixed $z \neq 0$, write $z = r\zeta$ with $r = \|z\|$ and $\zeta \in S^3$. The pushforward of Haar measure under $k \mapsto k\zeta$ is normalized surface measure on S^3 . By (3.4) and Corollary 3.2,

$$\int_{SU(2)} \mathcal{P}(kz)^m dk = r^{8m} \int_{S^3} p^m d\sigma = 0,$$

and

$$\begin{aligned} \int_{SU(2)} \mathcal{Q}(kz)^s \mathcal{P}(kz)^m dk &= r^{8m+2s} c_m \binom{m-1}{s-1} \\ &= a(z)^{4m+s} c_m \binom{m-1}{s-1}. \end{aligned}$$

The same formulas hold at $z = 0$. Substitution into (4.4) proves (4.1) and (4.2). If μ is not concentrated at the origin, then a^{4m+s} is nonnegative and positive on a set of positive μ -measure for every $m, s \geq 1$. For $1 \leq s \leq m$, combine this with $c_m > 0$ and $\binom{m-1}{s-1} > 0$ to obtain (4.3). $\square$

5 Root- $SU(2)$ transfer to compact simple groups

We first treat a compact connected simply connected simple Lie group $\tilde{G}$. Fix a maximal torus and a system of simple roots. For a simple root α , let ϖ_α be the corresponding fundamental weight. Since $\tilde{G}$ is simply connected, the highest-weight theorem provides an irreducible representation

$$\rho : \tilde{G} \longrightarrow U(V) \quad (5.1)$$

with highest weight ϖ_α . We choose a $\tilde{G}$ -invariant Hermitian inner product on V , linear in the first variable.

Lemma 5.1 (Visible root doublet). *Let $e_0 \in V$ be a unit highest-weight vector. There exist*

- (i) *a root subgroup $K_\alpha \subseteq \tilde{G}$ isomorphic to $SU(2)$;*
- (ii) *a K_α -invariant two-dimensional subspace $W = \text{span}_{\mathbb{C}}\{e_0, e_1\} \subseteq V$;*

such that the representation of K_α on W is unitarily equivalent to the defining representation of $SU(2)$ on $\mathbb{C}^2$.

Proof. The defining property of the fundamental weight is

$$\langle \varpi_\alpha, \alpha^\vee \rangle = 1,$$

where the brackets here denote the canonical weight–coroot pairing. Restrict the representation to the $\mathfrak{sl}_2$ -subalgebra associated with α . The vector e_0 is an $\mathfrak{sl}_2$ -highest-weight vector of highest weight 1. Hence the cyclic $\mathfrak{sl}_2$ -submodule generated by e_0 is the irreducible highest-weight-one module. If F_α is a nonzero lowering operator, then

$$F_\alpha e_0 \neq 0, \quad F_\alpha^2 e_0 = 0,$$

and the two weights are ϖ_α and $\varpi_\alpha - \alpha$. Let e_1 be a normalization of $F_\alpha e_0$ and set $W = \text{span}_{\mathbb{C}}\{e_0, e_1\}$.

The compact real form of the root $\mathfrak{sl}_2$ integrates to a homomorphism $\phi_\alpha : SU(2) \rightarrow \tilde{G}$. On W , the representation $\rho \circ \phi_\alpha$ is the irreducible highest-weight-one representation of $SU(2)$, hence the defining two-dimensional representation. This representation is faithful, so ϕ_α is injective. Thus $K_\alpha = \phi_\alpha(SU(2)) \simeq SU(2)$, and the image of K_α in $U(W)$ is the full special unitary group $SU(W)$. In particular, in every orthonormal coordinate system on W this image is the standard subgroup $SU(2) \subset U(2)$; no basis-invariance of the polynomial pair itself is required. These standard root-subgroup and highest-weight facts are treated in [2, Chapters V–VI] and [5, Parts II–III]. $\square$

Let $\Pi_W : V \rightarrow W$ be orthogonal projection. Since ρ is unitary and W is K_α -invariant, $W^\perp$ is also K_α -invariant, and therefore

$$\Pi_W \rho(k) = \rho(k) \Pi_W \quad (k \in K_\alpha). \quad (5.2)$$

Define

$$\Phi : \tilde{G} \longrightarrow W, \quad \Phi(g) = \Pi_W \rho(g) e_0. \quad (5.3)$$

Using the orthonormal basis (e_0, e_1) , identify W unitarily with $\mathbb{C}^2$. By Lemma 5.1 and the preceding observation, the image of K_α is the standard $SU(2)$ in these coordinates. Write

$$\Phi(g) = z_0(g) e_0 + z_1(g) e_1, \quad z_j(g) = \langle \rho(g) e_0, e_j \rangle. \quad (5.4)$$

Lemma 5.2 (Radial Haar pushforward). *The pushforward $\mu = \Phi_*(dg)$ of normalized Haar measure on $\tilde{G}$ is a compactly supported $SU(2)$ -invariant probability measure on $\mathbb{C}^2$, and it is not concentrated at the origin.*

Proof. For $k \in K_\alpha$, equations (5.2) and (5.3) give

$$\Phi(kg) = \rho(k)\Phi(g). \quad (5.5)$$

Left invariance of Haar measure therefore implies invariance of μ under the defining $SU(2)$ -action on $W \simeq \mathbb{C}^2$.

Since $\rho(g)e_0$ is a unit vector and Π_W is an orthogonal projection, $\|\Phi(g)\| \leq 1$; hence μ is supported in the closed unit ball. At the identity, $\Phi(1) = e_0$. By continuity, $\|\Phi(g)\| > 1/2$ on some nonempty open neighborhood of the identity. Every nonempty open subset of a compact group has positive Haar measure, so $\mu(\mathbb{C}^2 \setminus \{0\}) > 0$. $\square$

Apply (3.1) and (3.3) to the coordinate pair $(z_0(g), z_1(g))$. This defines functions A, u, v, τ, P, Q on $\tilde{G}$:

$$A = a \circ \Phi = |z_0|^2 + |z_1|^2, \quad P = \mathcal{P} \circ \Phi, \quad Q = \mathcal{Q} \circ \Phi. \quad (5.6)$$

Theorem 5.3 (Simply connected simple groups). *Let $\tilde{G}$ be a compact connected simply connected simple Lie group. The functions A, P, Q in (5.6) belong to $\mathcal{R}(\tilde{G})$, with $A \geq 0$ and $A \not\equiv 0$, and satisfy, for every $m \geq 1$,*

$$\int_{\tilde{G}} P(g)^m dg = 0. \quad (5.7)$$

For every integer $s \geq 1$,

$$\int_{\tilde{G}} Q(g)^s P(g)^m dg = c_m \binom{m-1}{s-1} \int_{\tilde{G}} A(g)^{4m+s} dg. \quad (5.8)$$

The right-hand side is strictly positive for $1 \leq s \leq m$ and is zero for $s > m$.

Proof. Each z_j in (5.4) is a matrix coefficient of ρ . By Lemma 2.2, its conjugate and every polynomial in the z_j and $\bar{z}_j$ are representative functions. Hence $A, P, Q \in \mathcal{R}(\tilde{G})$. Moreover, $A = \|\Phi\|^2 \geq 0$, and $A(1) = 1$. Lemma 5.2 allows us to apply Theorem 4.1 to $\mu = \Phi_*(dg)$, which gives all displayed identities and the strict positivity. $\square$

Proposition 5.4 (Closed forms in the defining representations). *Choose the defining representation and the first simple root for $SU(n)$ or $Sp(n)$. Write $(x)_k = x(x+1) \cdots (x+k-1)$ for the rising factorial. For $SU(n)$ with $n \geq 2$,*

$$\mathcal{I}_{SU(n)}(Q^s P^m) = c_m \binom{m-1}{s-1} \frac{(2)_{4m+s}}{(n)_{4m+s}} \quad (s \geq 1). \quad (5.9)$$

For $Sp(n)$ with $n \geq 1$,

$$\mathcal{I}_{Sp(n)}(Q^s P^m) = c_m \binom{m-1}{s-1} \frac{(2)_{4m+s}}{(2n)_{4m+s}} \quad (s \geq 1). \quad (5.10)$$

In particular, the first marked moments for $m = 1, 2, 3$ are

G	$m = 1$	$m = 2$	$m = 3$
$SU(2)$	$\frac{2}{3}$	$\frac{8}{15}$	$\frac{16}{35}$
$SU(3)$	$\frac{21}{4}$	$\frac{165}{4}$	$\frac{525}{2}$
$SU(4), Sp(2)$	$\frac{1}{14}$	$\frac{1}{165}$	$\frac{1}{175}$

Proof. For the defining representation of $SU(n)$, the map $\rho(g)e_0$ is the first column of g , uniformly distributed on the unit sphere in $\mathbb{C}^n$, and $A = |g_{11}|^2 + |g_{21}|^2$. If $n \geq 3$, the vector of squared coordinate moduli has the Dirichlet distribution with all parameters equal to 1; this follows, for example, by normalizing independent standard complex Gaussian variables. Hence

$$A \sim \text{Beta}(2, n-2), \quad \mathcal{I}_{SU(n)}(A^k) = \frac{B(k+2, n-2)}{B(2, n-2)} = \frac{(2)_k}{(n)_k}.$$

At the endpoint $n = 2$, one has $A \equiv 1$, and the same rising-factorial formula remains valid because $(2)_k/(2)_k = 1$. Substitution of $k = 4m + s$ into (5.8) proves (5.9).

The defining action of $Sp(n)$ is transitive on the unit sphere in $\mathbb{C}^{2n}$, and its invariant probability measure there is normalized surface measure. For $n \geq 2$, the same argument gives $A \sim \text{Beta}(2, 2n-2)$. At the endpoint $n = 1$, one has $Sp(1) = SU(2)$ and $A \equiv 1$, while $(2)_k/(2n)_k = 1$. This proves (5.10). The displayed values follow by direct simplification. $\square$

We now verify descent through the full center, including finite-type status on the quotient.

Lemma 5.5 (Center descent). *Let $Z = Z(\tilde{G})$. The functions A, u, v, τ, P, Q above are invariant under left multiplication by Z and are pullbacks of representative functions on the adjoint quotient $\tilde{G}/Z$. Consequently, they descend as representative functions to every central quotient $\tilde{G}/C$ with $C \leq Z$.*

Proof. By irreducibility and Schur's lemma, for every $c \in Z$ there is a unit scalar $\chi(c)$ such that $\rho(c) = \chi(c)I$. Therefore

$$z_j(cg) = \chi(c)z_j(g), \quad \overline{z_j(cg)} = \overline{\chi(c)}\overline{z_j(g)}.$$

Every quantity in (3.1), and hence A, P, Q , is a polynomial in the phase-balanced products $z_i \overline{z_j}$; all are therefore Z -invariant.

To see directly that the descended functions are representative, observe that $z_i \overline{z_j}$ is a matrix coefficient of $\rho \otimes \bar{\rho}$. The center acts on this tensor product as $\chi(c)\overline{\chi(c)}I = I$, so $\rho \otimes \bar{\rho}$ factors through $\tilde{G}/Z$. The functions A, u, v, τ are linear combinations of its matrix coefficients, while P and Q are matrix coefficients of finite direct sums of tensor powers thereof. Thus all are representative functions on $\tilde{G}/Z$, and their pullbacks to every intermediate central quotient remain representative. $\square$

Corollary 5.6 (Every compact simple central form). *Let H be a compact connected Lie group with simple Lie algebra. Then there exist $A_H, P_H, Q_H \in \mathcal{R}(H)$, with $A_H \geq 0$ and $A_H \not\equiv 0$, such that for every $m \geq 1$,*

$$\mathcal{I}_H(P_H^m) = 0,$$

and, for every $s \geq 1$,

$$\mathcal{I}_H(Q_H^s P_H^m) = c_m \binom{m-1}{s-1} \mathcal{I}_H(A_H^{4m+s}).$$

In particular, H does not have the Mathieu property.

Proof. There is a compact connected simply connected simple group $\tilde{G}$ and a subgroup $C \leq Z(\tilde{G})$ such that $H \simeq \tilde{G}/C$. By Lemma 5.5, the triple from Theorem 5.3 descends to H . The quotient map pushes normalized Haar measure forward to normalized Haar measure, so the moment identities and positivity are unchanged by Lemma 2.3. $\square$

6 Arbitrary compact connected groups

The remaining structural step is to exhibit a simple quotient of every nonabelian compact connected group.

Proposition 6.1 (Adjoint simple quotient). *Let G be a nonabelian compact connected Lie group. Then there is a continuous surjective homomorphism*

$$\pi : G \twoheadrightarrow H$$

onto a compact connected adjoint simple Lie group H .

Proof. Let $\mathfrak{g}$ be the Lie algebra of G . Compactness gives a decomposition into ideals

$$\mathfrak{g} = \mathfrak{z} \oplus \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_r, \tag{6.1}$$

where $\mathfrak{z}$ is the center and the $\mathfrak{s}_i$ are compact simple Lie algebras. Since G is nonabelian, $r \geq 1$.

The adjoint action of the connected group G cannot nontrivially permute the finite set of simple ideals, so each $\mathfrak{s}_i$ is $\text{Ad}(G)$ -invariant. Fix i and define

$$\pi_i : G \longrightarrow \text{Aut}(\mathfrak{s}_i), \quad \pi_i(g) = \text{Ad}(g)|_{\mathfrak{s}_i}.$$

The image is connected, hence lies in the identity component $\text{Int}(\mathfrak{s}_i)$ of $\text{Aut}(\mathfrak{s}_i)$. The differential of π_i is

$$d\pi_i(X) = \text{ad}(X)|_{\mathfrak{s}_i}.$$

Using (6.1), its image is exactly $\text{ad}(\mathfrak{s}_i)$, the Lie algebra of $\text{Int}(\mathfrak{s}_i)$. Because $\pi_i(G)$ is compact, it is closed. A closed connected subgroup with the same Lie algebra as the connected group $\text{Int}(\mathfrak{s}_i)$ is the whole group. Thus π_i is surjective onto the compact connected adjoint simple group $H = \text{Int}(\mathfrak{s}_i)$. $\square$

Proof of Theorem 1.2. By Proposition 6.1, choose a surjection $\pi : G \twoheadrightarrow H$ onto an adjoint compact simple group. Choose $A_H, P_H, Q_H \in \mathcal{R}(H)$ as in Corollary 5.6. Set

$$A = A_H \circ \pi, \quad P = P_H \circ \pi, \quad Q = Q_H \circ \pi.$$

Lemma 2.3 gives (1.1) and (1.2), and preserves the pointwise nonnegativity and nonvanishing of A . $\square$

7 The torus direction and completion of the classification

Let T be a compact connected abelian Lie group. Then T is a torus and its character lattice $\hat{T}$ is isomorphic to $\mathbb{Z}^d$ for some $d \geq 0$. Under the resulting identification

$$\mathcal{R}(T) \simeq \mathbb{C}[x_1^{\pm 1}, \dots, x_d^{\pm 1}],$$

normalized Haar integration is Laurent constant-term extraction.

We use the following theorem of Duistermaat and van der Kallen [4].

Theorem 7.1 (Duistermaat–van der Kallen). *Let $0 \neq f \in \mathbb{C}[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$. If $\text{CT}(f^m) = 0$ for every $m \geq 1$, then the origin does not belong to the convex hull of the exponent support of f .*

Corollary 7.2 (The torus Mathieu property). *Every torus has the Mathieu property.*

Proof. Let $f, h \in \mathbb{C}[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$ and assume $\text{CT}(f^m) = 0$ for every $m \geq 1$. If $f = 0$ or $h = 0$, the conclusion is immediate. Assume $f \neq 0$ and $h \neq 0$. By Theorem 7.1, a real linear functional ℓ on $\mathbb{R}^d$ strictly separates the finite support of f from the origin. Thus there is $\delta > 0$ such that $\ell(\alpha) \geq \delta$ for every exponent α in the support of f . If C is the minimum of ℓ on the finite support of h , then every exponent in the support of hf^m has ℓ -value at least $C + m\delta$. For all sufficiently large m this is positive, so the zero exponent is not in the support of hf^m . Hence $\text{CT}(hf^m) = 0$ for all sufficiently large m . $\square$

Proof of Theorem 1.1. For compact connected Lie groups, abelianity is equivalent to being a torus. Corollary 7.2 proves that every torus has the Mathieu property. If G is nonabelian, Theorem 1.2 produces $P, Q \in \mathcal{R}(G)$ with $P^m \in \ker \mathcal{I}_G$ for every $m \geq 1$ but $QP^m \notin \ker \mathcal{I}_G$ for any $m \geq 1$. Thus $\ker \mathcal{I}_G$ is not a Mathieu subspace. This proves all equivalences. $\square$

8 The abelian reductions and an explicit transformed witness

The universal conjectures of Müger–Tuset are already false indirectly: by [10, Remark 6.7(2)] they imply the Jacobian conjecture, so the three-dimensional Jacobian counterexample discussed in the introduction disproves them by contraposition. There is also a direct admissible witness obtained immediately from [7, Theorem 2.1]. Write

$$f_0(t, w) = (1 - w^{-1})((1 - t) + tw), \quad \tilde{f}(x, w) = f_0(x^2, w).$$

The substitution $t = x^2$ and the moment identities in [7] give, for every $m \geq 1$,

$$\begin{aligned} \int_0^1 \text{CT}_w(\tilde{f}(x, w)^m) x \, dx &= 0, \\ \int_0^1 \text{CT}_w(w^{-1} \tilde{f}(x, w)^m) x \, dx &= \frac{(-1)^{m-1}}{2(m+1)} \neq 0. \end{aligned}$$

Thus $\tilde{f}$, w^{-1} , and the admissible weight $\delta(x) = x$ already give a direct counterexample to Conjecture 6.3 at $N = M = 1$; the unchanged w -spectrum $\{-1, 0, 1\}$ also contradicts Conjecture 6.6.

The purpose of this section is different. We identify the exact image of the universal Hopf pair under the square-root-free Müger–Tuset coordinates for $SU(2)$. The resulting witness lies in the distinguished group-coordinate algebra, uses the actual $SU(2)$ weight $\delta(x) = x$, and retains the full positive Pascal marker tower.

Proposition 8.1 (An explicit $SU(2)$ Laurent witness). *Let $x \in [0, 1]$ and let w be a unit-circle variable. Define*

$$U = 2x(1 - x^2)w, \quad V = 2xw^{-1}, \quad T = 1 - 2x^2,$$

and

$$P_{\text{ab}} = (1 + U)(V - (2 + U)T^2), \quad Q_{\text{ab}} = U. \tag{8.1}$$

Then, for every $m \geq 1$,

$$2 \int_0^1 \text{CT}_w(P_{\text{ab}}^m) x \, dx = 0, \tag{8.2}$$

and, for every $s \geq 1$,

$$2 \int_0^1 \text{CT}_w(Q_{\text{ab}}^s P_{\text{ab}}^m) x \, dx = c_m \binom{m-1}{s-1}. \tag{8.3}$$

Moreover,

$$\begin{aligned} P_{ab} = & 2xw^{-1} - 2(6x^4 - 6x^2 + 1) \\ & + 6x(x^2 - 1)(2x^2 - 1)^2w \\ & - 4x^2(x^2 - 1)^2(2x^2 - 1)^2w^2, \end{aligned} \quad (8.4)$$

so its w -spectrum is exactly

$$\mathrm{Sp}_w(P_{ab}) = \{-1, 0, 1, 2\}. \quad (8.5)$$

Proof. One has $UV + T^2 = 1$. Consequently,

$$UP_{ab} = (1 + U)(1 - (1 + U)^2T^2),$$

which is the same defect-one identity as (3.9). The change of variables $t = 1 - 2x^2$ gives $2x \, dx = -\frac{1}{2} \, dt$. Constant-term extraction in w , followed by evenness in t , therefore reduces (8.2) and (8.3) exactly to the calculation in Theorem 3.1. Direct expansion gives (8.4); each of its four displayed coefficient polynomials is nonzero, proving (8.5).

We now identify this pair with the M\"uger–Tuset transform without retaining any conjugate variables. Write

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SU(2).$$

Since $\bar{a} = d$ and $\bar{c} = -b$ on $SU(2)$, the Hopf quantities of the first column have the following polynomial representatives in $\mathbb{C}[a, b, c, d]$:

$$A_0 = ad - bc, \quad U_0 = -2ab, \quad V_0 = 2cd, \quad T_0 = ad + bc. \quad (8.6)$$

Thus the restrictions to $SU(2)$ of the universal pair are represented by

$$\hat{P} = (A_0 + U_0)(A_0^2V_0 - (2A_0 + U_0)T_0^2), \quad \hat{Q} = U_0, \quad (8.7)$$

which are genuine polynomials in the four matrix entries.

Multiplication by the maximal-torus factor $\psi(z) = \mathrm{diag}(z, z^{-1})$ sends

$$(a, b, c, d) \longmapsto (az, bz^{-1}, cz, dz^{-1}).$$

Every expression in (8.6) is unchanged, so the variable z cancels. Under the square-root-free substitution of [10, Lemma 5.2],

$$(a, b, c, d) = (iw(1 - x^2), ix, ix, -iw^{-1}),$$

one obtains

$$A_0 \longmapsto 1, \quad U_0 \longmapsto U, \quad V_0 \longmapsto V, \quad T_0 \longmapsto T.$$

Hence (8.7) maps exactly to (8.1). This is the polynomial-entry substitution required by M\"uger–Tuset: no conjugation variable survives. In the proof of Proposition 5.3 of [10], they also note that the substituted polynomial is independent of the chosen polynomial representative. After normalizing the circle integrals as constant terms and Haar measure to have total mass one, their Proposition 5.3 gives precisely the functional $2 \int_0^1 \mathrm{CT}_w(\cdot) x \, dx$, in agreement with (8.2) and (8.3). $\square$

Corollary 8.2 (Failure of the universal abelian conjectures). *The universal moment conjecture and the universal convex-support conjecture of Müger–Tuset [10, Conjectures 6.3 and 6.6] are false. The stronger growth assertion proposed in [10, Remark 6.7(1)] is also false. More generally, every auxiliary abelian conjecture in [14, 15, 16] that is proved there to imply the Mathieu conjecture for a specified nonabelian compact connected group is false for that group.*

Proof. Up to the harmless overall factor 2, the functional in Proposition 8.1 is the Müger–Tuset functional $\int_{(1,1)}(\cdot)x$. Taking $f = P_{\text{ab}}$, $g = Q_{\text{ab}}$, and $\delta(x) = x$, equations (8.2) and (8.3) with $s = 1$ give vanishing pure moments but nonzero g -marked moments for every $m \geq 1$. This directly disproves Conjecture 6.3. Equation (8.5) gives $0 \in \text{Sp}_w(P_{\text{ab}})$, hence directly disproves Conjecture 6.6. At the same time,

$$\limsup_{m \rightarrow \infty} \left| 2 \int_0^1 \text{CT}_w(P_{\text{ab}}^m)x \, dx \right|^{1/m} = 0,$$

which contradicts the stronger positive-growth assertion in Remark 6.7(1). The statements in Zwart’s papers follow by contraposition from the implication theorems proved there and Theorem 1.1. $\square$

Remark 8.3 (The mixed case $N = M = 1$). *For the actual $SU(2)$ reduction, Müger–Tuset use $N = 1$, $M = 2$, and $\delta(x) = x$; the additional Laurent variable is the maximal-torus variable z . The pair in Proposition 8.1 lies in their algebra $\mathcal{A}(SU(2))$ and is independent of z . Deleting this unused variable places it at $N = M = 1$, the smallest case containing both a polynomial variable and a Laurent variable. The quadratic reparametrization at the beginning of this section gives another admissible witness at the same pair (N, M) . What distinguishes Proposition 8.1 is its origin as the exact transform of the universal Hopf pair and its complete positive Pascal marker tower. By [10, Remark 6.4], the case $N = 0$ is the torus theorem of Duistermaat–van der Kallen, while only partial results were available for $N = 1$, $M = 0$.*

9 Further remarks

Remark 9.1 (Strength of the failure). *For every nonabelian compact connected G , the failure is uniform in the exponent: the same fixed multiplier Q detects every positive power of P . The full Pascal marker tower (1.2) holds on G itself, not merely on a simply connected simple cover.*

Remark 9.2 (No classification by Lie type). *The proof uses only one simple root and its highest-weight-one doublet. It therefore handles the classical and exceptional Lie types uniformly, as well as all simply connected, adjoint, and intermediate central forms. Products, torus extensions, and finite central identifications are absorbed by the adjoint-simple-quotient argument.*

Remark 9.3 (Scope). *The theorem classifies Mathieu’s original compact connected group problem. It does not by itself classify disconnected compact groups, noncompact groups, or homogeneous spaces G/H with H nonnormal; those require separate formulations or additional transfer arguments.*

AI-assisted research disclosure

The author used ChatGPT 5.6 Sol Pro (OpenAI) and Claude Opus 5 (Anthropic) for exploratory calculations, proof organization, literature discovery, and language editing. The author reviewed and checked the mathematical arguments, calculations, citations, and final text and assumes sole responsibility for the contents. Neither AI system is an author.

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